

Distortion of cortical hue geometry in color vision deficiency enables an individualized  
color correction filter

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## Abstract

Color vision deficiency (CVD) is typically corrected with filters that target a population-average retina and treat every user alike. Yet the cortical color representation that perception reads out is reshaped individually, and whether it could guide a personalized correction is unknown. Here we show, using fMRI in two adults with CVD studied as single cases alongside healthy controls, that the cortical representation of color is not reduced in overall signal but distorted in its response structure. Both CVD participants discriminated all displayed hues across the visual hierarchy, yet the structure of their color representation was distorted in two complementary ways. Continuous interpolation between hues failed in area hV4, and the geometry of color representations itself was altered in early visual cortex (V1 and V2), differing between the two individuals. We then summarized each person's distortion with a compact two-component cortical model and analytically inverted it into a per-individual stimulus-space filter, designed to move the CVD color perception and cortical representation structure toward that of a healthy observer. In a second session in the deutan participant, the personalized filter moved the impaired cortical hue geometry back toward the healthy-control reference, where a deployed accessibility filter did not; both filters restored behavioral discrimination comparably, and validation in further individuals remains open. These findings recast CVD as an individually patterned distortion of cortical color structure rather than a loss of chromatic signal, one that can be inverted into a personalized correction.

*Keywords:* color vision deficiency, anomalous trichromacy, fMRI, hV4, representational similarity analysis, forward encoding model, personalized color correction

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## Introduction

Color vision deficiency (CVD) affects approximately 8% of men and 0.4% of women of European ancestry (Birch, 2012). It arises from altered spectral sensitivity of the L- or M-cone photopigment. This altered sensitivity is driven by X-linked polymorphisms in the cone opsin genes (Deeb, 2005; Neitz & Neitz, 2011). In anomalous trichromats, the deviant cone peak is displaced by roughly 2–12 nm from its normal position (Stockman & Sharpe, 2000). This small displacement reduces the L–M cone-opponent signal rather than abolishing it.

A weakened L–M signal yields a characteristic peripheral phenotype. Affected observers show reduced red–green discrimination, frequent category confusions, and elevated just-noticeable differences along the L–M chromatic axis. The Ishihara test screens for the resulting red–green confusions (Ishihara, 1917). The elevated discrimination thresholds are quantified instead by threshold psychophysics (Boehm, MacLeod & Bosten, 2014; Bosten, 2019).

Current CVD correction methods fall into two structurally related families. Hardware filters form the first family. Dichroic notch lenses, such as the EnChroma line, are designed to filter narrow spectral bands and amplify the residual red–green signal. Software filters form the second. Brettel–Viénot–Mollon simulation (Brettel, Viénot & Mollon, 1997) and the parameterized Machado retinal model (Machado, Oliveira & Fernandes, 2009) forward-project a stimulus onto an anomalous retina. Their Daltonization inverses, including recent algorithmic reformulations (Akalin et al., 2025), redistribute the lost contrast into channels the user can still see. Across this diversity, the two families share one commitment: they optimize against a population-average retina and apply the same transform to every user.

Attempts to escape this average-user ceiling have mostly taken one route. They tune a retinal cone-shift parameter to a user’s responses on plate-style or appearance-matching tests. The criterion is then the appearance *report*, not the cortical

representation that downstream tasks read out. A ceiling follows, because the phenotype varies markedly even within one diagnostic category (Bosten, 2019). On a clinical chromatic-discrimination test, color-enhancing glasses improved discrimination for only one of two consumer products tested (Patterson et al., 2022).

Population-average filters likewise shift the appearance of color while leaving generalized color discrimination largely unchanged (Somers, Franklin & Bosten, 2024). What is missing is a filter derived from, and grounded in, the user’s *own* neural color geometry.

CVD is not an on–off loss of color. The downstream cortical color machinery stays intact. It receives an input that is quantitatively reweighted but qualitatively similar to a normal input. The cortical color representation is therefore not absent but structurally distorted. These features make the cortical representation the natural level at which to examine the distortion.

Cortical color is a distributed, hierarchical computation (Gegenfurtner, 2003; Shapley & Hawken, 2011; Conway, Eskew, Martin & Stockman, 2018). Early visual cortex carries cone-opponent signals. Higher-order regions, in particular hV4, support a population code for hue (Brouwer & Heeger, 2009), and act as a hub linking seen and remembered color (Bannert & Bartels, 2018). Continuous hue can be reconstructed from these distributed patterns (Brouwer & Heeger, 2009), and hue-selective responses to intermediate colors appear as early as V1 (Parkes, Marsman, Oxley, Goulernas & Wuerger, 2009; Kuriki, Sun, Ueno, Tanaka & Cheng, 2015). This code is shared enough across people that one observer’s color responses can be read out in a common space, an approach recently validated in healthy observers (Bannert & Bartels, 2025).

Anomalous trichromats do more than lose sensitivity. They also compensate. Long-term cortical adaptation reshapes hue appearance independently of the retina (Boehm et al., 2014; Webster, 2015; Tregillus et al., 2021). In deutan observers, chromatic-contrast adaptation exceeds what threshold sensitivity loss predicts, consistent with a post-receptoral gain that partly offsets the weaker L–M signal (Basim, Goddard, Yang & Webster, 2025). Chromatic responses also realign toward the blue–yellow direction in these observers (Emery, Volbrecht, Peterzell & Webster, 2021).

Because this evidence comes from a separate paradigm, it motivates rather than establishes a cortical locus, and we do not replicate it here. Together these findings locate key variance at the retinal–cortical interface, not at the retina alone.

Because this compensation lives in cortex, a correction calibrated to the retina or to appearance reports inherits it and acts at the wrong level. We therefore take the individual’s cortical readout, not the simulated retina, as the reference for correction. This is a design hypothesis, and the rest of the paper tests it.

Three concrete gaps remain. (*Gap 1*) Neural studies of color describe geometry at the group level in healthy observers (Brouwer & Heeger, 2009; Kuriki et al., 2015; Bannert & Bartels, 2018). The cortical color geometry of an individual CVD observer, the personal trait on which a tailored correction would be built, has not been characterized. (*Gap 2*) The measures applied to CVD do not capture this geometry. fMRI in CVD has quantified response magnitude: contrast-response gain in anomalous trichromacy (Tregillus et al., 2021), and activation in dichromacy and achromatopsia (Rina et al., 2024). Where multivoxel patterns have been analyzed, the readout has been classification accuracy. Magnitude indexes signal strength and accuracy indexes category identity, but neither preserves the continuous relational structure (Brouwer & Heeger, 2013) that a hue-space filter must reshape. (*Gap 3*) Across studies that analyze multivoxel color patterns, we are not aware of a corrective filter inverted from an individual’s own cortical representation. Whether such a high-dimensional signal can be turned into a realizable, individualized correction is untested.

We ask whether an individual’s cortical color geometry can serve as the reference for a personalized correction. To answer this, we characterize two adults with CVD, one with each major red–green subtype, against a reference group of healthy controls (see Methods). Observations from the single protan participant are exploratory and need independent replication.

A stimulus-space filter is both warranted and tractable only in a specific regime. The displayed colors must stay neurally distinguishable, or the filter has no signal to act

on. The continuous arrangement of hues must also be distorted, or there is nothing to correct. Our questions follow this logic.

1. **Describe.** Does an individual’s cortical hue geometry in CVD carry structured distortion, with categorical discrimination preserved but the continuous arrangement of hues impaired?
2. **Summarize.** Can that distortion be captured by a few interpretable parameters, treated as a compact description rather than as physiological measurements?
3. **Correct.** Can a stimulus-space correction be derived from those parameters as a realizable, individualized filter?
4. **Validate.** Does the filter move the cortical representation toward the healthy-control reference and outperform a deployed accessibility filter?

We expect the continuous arrangement of hues to be selectively impaired while categorical discrimination is preserved, with the largest distortion among the blue-to-magenta hues. A few interpretable parameters should capture this distortion, and inverting them should yield a realizable, individualized correction.

Together, these steps yield a filter built for one individual. Its input is the cortical signal, its parameters are the fitted description, and its output is a stimulus shift. Each is defined at the level of the single observer, the level at which any correction must ultimately act.

## Methods

The analysis follows three stages. First, two neural measures characterized each CVD participant’s hue-processing deficit: leave-one-color-out (LOCO) decoding at hV4 yielded a per-hue vulnerability profile (§), and the between-group RDM difference ( $\Delta$ RDM) captured the per-pair geometric deviation of CVD representational geometry from the HC mean (§). Second, two candidate distortion model classes were fit by minimizing a combination of neural and behavioural losses, with parameters selected by

a pre-specified stability procedure (§-§). Third, the fitted distortion was analytically inverted to yield a per-subject stimulus-space correction filter (§); Figure 3 summarizes this fit-and-invert workflow. This filter was then evaluated in a repeat fMRI session and a 2AFC behavioural discrimination test (§).

## Participants

Thirteen volunteers were recruited under Institutional Review Board approval (#2510/002-023) and provided written informed consent. Color vision was screened using the 14-plate Ishihara pseudoisochromatic test (Ishihara, 1917). Participants identifying  $\geq 9$  plates correctly were classified as healthy controls (HC;  $n = 7$ , 3 female, age  $22.7 \pm 2.5$  years). Two male participants scoring below this threshold were classified as CVD: sub-08 (deuteranomalous, 5/14 plates corrected) and sub-09 (protanomalous, 7/14 plates corrected). Four additional participants were excluded for failing to complete the full scanning protocol. All CVD results are interpreted as single-case demonstrations using the Crawford and Howell (1998) modified  $t$ -test; no population-level inference is made.

## Stimuli and task

Eight isoluminant chromatic stimuli were drawn from CIE  $L^*a^*b^*$  space at equal angular spacing ( $0^\circ$ – $315^\circ$ ,  $45^\circ$  steps;  $L^* = 75$ , chroma = 40) (Commission Internationale de L’Eclairage, 1986), spanning red, orange, yellow, green, cyan, blue, purple, and magenta (Figure 1A). A gray filler ( $a^* = b^* = 0$ ) served as a null condition.

Stimuli were presented in a Rapid Serial Visual Presentation (RSVP) design to sustain attention and fixation. Each stimulus was a uniform circular chromatic patch (a colored disc, 1.5 s per stimulus) shown in pseudo-random order optimized with Neurodesign (Durnez, Moerkerke & Nichols, 2018). Participants pressed a button upon detecting two consecutive ‘K’s within a rapid serial letter stream presented at the centre of the chromatic disc. Each run contained eight repetitions of all eight chromatic stimuli and the gray filler; inter-stimulus intervals were drawn from  $\{3, 4.5, 6\}$  s. Each

session comprised six runs ( $\approx 7$  min each). The experiment was coded in PsychoPy 2022.2.5 (Peirce et al., 2019).

All participants completed JND behavioral testing in the same scanning session as the main MRI acquisition. The deutan participant (sub-08) additionally completed a second session in which JND was repeated under the per-subject stimulus-space filter, alongside a second fMRI acquisition (see the Behavioral tasks section).

## MRI acquisition and preprocessing

Data were acquired on a Siemens 3T MAGNETOM Cima.X scanner with BioMatrix 3T coils. To constrain coverage to posterior occipital cortex we used 24 oblique slices perpendicular to the calcarine sulcus (Brouwer & Heeger, 2009) (TR 1.5 s, TE 30 ms, FA  $75^\circ$ ,  $2 \times 2 \times 2$  mm,  $96 \times 80$  matrix).

Data were converted to BIDS format and defaced with ezBIDS (Levitas et al., 2024). Functional images were registered to the T1w image via mutual-information coregistration (FreeSurfer `mri_coreg`; Wells, Viola, Atsumi, Nakajima & Kikinis, 1996; Maes, Collignon, Vandermeulen, Marchal & Suetens, 1997). T1w-to-MNI normalization used 12-DOF affine (FSL FLIRT; Jenkinson, Bannister, Brady & Smith, 2002) followed by nonlinear warping (FNIRT; Andersson, Jenkinson & Smith, 2007), yielding 2 mm isotropic MNI-space BOLD data.

## ROI definition and response estimation

Bilateral V1, V2, V3, and hV4 were defined from the Wang probabilistic atlas (Wang, Mruczek, Arcaro & Kastner, 2015) at  $> 50\%$  probability, intersected with each participant’s BOLD brain mask (voxel counts: V1  $655 \pm 214$ ; V2  $451 \pm 145$ ; V3  $103 \pm 29$ ; hV4  $63 \pm 22$ ; mean  $\pm$  SD).

Single-trial response amplitudes were estimated with a two-stage GLM (Dale, 1999; Brouwer & Heeger, 2009, 2013). First, an ROI-level hemodynamic response function (HRF) was extracted from a finite-impulse-response (FIR) model (8-TR basis set). Voxels in the top 50% of variance explained by the mean ROI HRF were retained (V1  $328 \pm 107$ ; V2  $226 \pm 73$ ; V3  $52 \pm 15$ ; hV4  $32 \pm 11$  voxels). Second, the estimated



HRF was convolved with per-color design matrices and fitted by ridge regression to yield one amplitude per (voxel, color, run), organized as a  $6 \times 8 \times V$  array per subject.

Voxel amplitudes were aligned across runs using within-subject Procrustes rotation (Gower, 1975): runs 2–6 were mapped to run 1 via the orthogonal transformation minimizing  $\|X_{\text{ref}} - XQ\|_F$  (scaling prohibited).

### Shared Response Model

Cross-subject alignment and dimensionality reduction were performed with the Shared Response Model (SRM; Chen et al., 2015). SRM estimates per-participant orthonormal mappings  $W_i \in \mathbb{R}^{V \times k}$  ( $V$  voxels,  $k$  latent dimensions) and a group-shared response matrix  $S \in \mathbb{R}^{k \times 8}$ :

$$\min_{W_i, S} \sum_{i=1}^N \|X_i - W_i S\|_F^2 \quad \text{s.t.} \quad W_i^\top W_i = I_k \quad (1)$$

SRM was trained exclusively on the seven HC participants to avoid circular inference in CVD comparisons. Each CVD participant was subsequently projected into the fixed HC-derived space by solving  $W_{\text{CVD}} = UV^\top$  where  $U\Sigma V^\top = \text{SVD}(X_{\text{CVD}} \cdot S^\dagger)$ . Reduced dimensions were chosen by mean-rank aggregation across seven-fold leave-one-subject-out (LOSO) cross-validation, using three SRM-quality metrics (RDM split-half reliability, cross-subject RDM correlation, and reconstruction error): V1  $k = 4$ ; V2  $k = 4$ ; V3  $k = 3$ ; hV4  $k = 3$ .

### Forward encoding model

Both discrimination (LORO) and interpolation (LOCO) decoding used the same forward encoding model (Brouwer & Heeger, 2009). The model represents each stimulus as a  $F = 6$ -dimensional channel vector  $\mathbf{c}(\theta) = [c_1(\theta), \dots, c_F(\theta)]^\top$ , where  $c_k(\theta) = \max(0, \cos(\theta - \theta_k))^2$  is a half-wave rectified squared cosine tuned to preferred hue  $\theta_k$  (preferred hues spaced equally at  $60^\circ$ ; Brouwer & Heeger, 2009). A weight matrix  $W \in \mathbb{R}^{F \times V}$  is estimated by ridge regression with the regularization parameter  $\alpha$  selected by generalized cross-validation (GCV; Golub, Heath & Wahba, 1979; see

Supplementary for implementation details) to predict voxel amplitudes from channel activations:  $\hat{X} = CW^\top$ . Decoding inverts the encoding direction: for a held-out voxel pattern  $\mathbf{x}$ , channel activations are reconstructed as  $\hat{\mathbf{c}} = W\mathbf{x}$ , and the candidate hue whose basis vector has the highest Pearson correlation with  $\hat{\mathbf{c}}$  is returned as the decoded color (Figure 2). Alternative decoders evaluated during model selection are described in Appendix .

### Evaluating color discrimination with cross-run decoding

Categorical discrimination was assessed with leave-one-run-out (LORO) cross-validation in SRM-aligned space using the forward encoding model (see the Forward encoding model section). For each of the six runs,  $W$  was estimated on the remaining five runs using mean-subtracted SRM projections, and each held-out run’s 8 color patterns were decoded by nearest-neighbour matching. Primary discrimination performance was the 8-class exact accuracy (chance =  $1/8 = 0.125$ ), averaged across six folds.

Group-level HC–CVD comparison used a Mann–Whitney  $U$  test on cross-subject generalization scores: the encoding model trained on each HC participant’s SRM-aligned data was applied to every other participant’s aligned patterns, yielding 21 unique HC-to-HC and 14 HC-to-CVD pairwise generalization scores. Within-ROI single-case tests used the Crawford and Howell (1998) modified  $t$ -test (one-tailed, lower).

### Evaluating color interpolation with cross-color decoding

Continuous hue interpolation was assessed with leave-one-color-out (LOCO) cross-validation using the same forward encoding model (see the Forward encoding model section). Each of the 8 hues was held out in turn;  $W$  was estimated on the remaining 7 hues using all 6 runs (pooled 42-sample design matrix), and the held-out hue was predicted by nearest-neighbour matching in channel-output space.

**Adjacent accuracy.** The primary cross-color decoding metric was adjacent accuracy: the proportion of predictions that fell within  $\pm 1$  hue step ( $45^\circ$ ) of the target hue, averaged across the 6 runs of each test color (range 0–1; chance level under a

uniform random predictor =  $3/8 = 0.375$ ; exact-accuracy chance =  $1/8 = 0.125$ ). For HC above-chance testing and for pairwise HC–CVD comparisons at hV4,  $p$ -values were obtained from 1,000 random color-label permutations and from the Crawford and Howell (1998) modified  $t$ -test (one-tailed, lower), respectively.

***Vulnerability profile.*** The 8-dimensional vector of per-hue adjacent-accuracy values,  $\mathbf{v} \in [0, 1]^8$ , constitutes each participant’s *hue vulnerability profile*. For group-level model fit, we report Spearman  $\rho$  between the distortion-model-predicted and observed  $\mathbf{v}$  (see the Inverse fitting section). Per-hue group comparisons (Figure 4C) used the Crawford and Howell (1998) modified  $t$ -test (one-tailed, uncorrected). Effect sizes are reported as Hedges’  $d$  (Hedges & Olkin, 1985) with the Crawford–Howell correction for small- $n$  comparisons.

### Between-group representational dissimilarity

To characterize the representational geometry of LOCO impairment in SRM-aligned space, we computed two measures. Mean pairwise disparity captures whether CVD geometry is globally more dispersed than HC. The between-group RDM difference,  $\Delta\text{RDM}$ , captures which color pairs drive the deviation and tests whether a cortical distortion model accounts for the pattern.

***Pairwise disparity.*** For each participant and ROI, the RDM was the  $8 \times 8$  matrix of pairwise correlation distances ( $1 - \text{Pearson } r$ ). Mean pairwise disparity was the average of the 28 upper-triangle elements. To provide an unbiased HC reference for the held-out HC participant, the HC disparity distribution was estimated by a leave-one-subject-out (LOO) procedure. Under the directional prediction that CVD geometry is more dispersed than HC, individual CVD disparity was compared to the HC LOO distribution by the Crawford and Howell (1998) modified  $t$ -test (one-tailed, upper). Significance was assessed at  $\alpha = 0.05$ .

***Geometric deviation from HC and model correspondence.*** The per-pair deviation of each CVD participant’s RDM from the HC mean was

$\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC}}$ , where  $\overline{\text{RDM}}_{\text{HC}}$  is the grand mean RDM over the seven HC participants. Correspondence between the observed  $\Delta\text{RDM}$  and the prediction of

the two-component distortion model (see Inverse fitting) was reported descriptively as the Spearman  $\rho$  between the upper-triangle elements of the two matrices.

### Candidate distortion models

Two candidate model classes were evaluated. Each parameterizes the angular distortion  $\delta\theta(\theta)$  that maps the displayed hue angle  $\theta$  to the angle perceived by a CVD observer.

***Candidate Model 1: retinal-plus-gain (R+C)..*** The prevailing account attributes CVD color distortion to a cone-spectral shift at the retinal level (Machado et al., 2009), optionally augmented by a cortical compensation gain  $g$  (Boehm et al., 2014; Tregillus et al., 2021):

$$\delta\theta_{RC}(\theta; g) = (2 - g) \delta\theta_{Mach}(\theta) \quad (2)$$

where  $\delta\theta_{Mach}$  is the per-hue angular shift predicted by Machado simulation applied to Stockman & Sharpe LMS cone fundamentals (Stockman & Sharpe, 2000). The gain  $g$  parameterizes cortical compensation: at  $g = 1$  the cortical stage leaves the retinal distortion unchanged; at  $g = 2$  it fully cancels the retinal shift; at  $g > 2$  cortical processing reverses the direction of the retinal shift, moving the perceived hue past the undistorted color. The confusion axis is the color-space direction along which CVD observers cannot distinguish hues from one another. Because  $\delta\theta_{Mach}$  lies entirely along this axis, R+C can only displace colors along this single direction and cannot account for distortions in other color directions.

***Candidate Model 2: two-component cortical (2-comp).*** We propose a parametric model of the hue distortion pattern measured in § and §:

$$\delta\theta(\theta; \beta_s, \beta_c) = \beta_s \cos(\theta - 90^\circ) + \beta_c \cos(\theta - \theta_{conf}) \quad (3)$$

where  $\theta_{conf}$  is the CVD-subtype confusion axis ( $16^\circ$  protan;  $150^\circ$  deutan), derived from Stockman & Sharpe cone fundamentals (Stockman & Sharpe, 2000).  $\beta_s$  parameterizes distortion along the S-cone axis ( $\theta = 90^\circ$ ) and  $\beta_c$  parameterizes distortion along the individual confusion axis; these two components span the axes of distortion identified by

the neural analysis (§, §). Both signs of  $\beta_c$  are unconstrained; the basis for constraining  $\beta_s \geq 0$  is detailed in Supplementary §S1. Model adequacy was assessed by the procedure in §; results are reported in §.

### Inverse fitting

Per-subject distortion parameters  $(\beta_s, \beta_c)$  were estimated by finding the distortion that, when applied to HC-derived neural representations, best reproduces the observed CVD responses. Three loss families served as the fitting objective.

**Behavioral JND ratio ( $L_\gamma$ ).** For each color pair, the JND ratio  $\gamma = \gamma_{\text{CVD}}/\bar{\gamma}_{\text{HC}}$  quantifies how much more difficult discrimination is for a CVD observer relative to the HC mean (§). Two variants enter the fitting: a subtype-targeted pair ratio  $\gamma_{\text{subtype}}$  (the hue pair on each participant’s confusion axis predicted to show the largest distortion) and an aggregate ratio  $\gamma_{\text{all}}$  averaged across all tested pairs.  $L_\gamma$  penalises rank discordance between the model-predicted distortion magnitude at each pair (from Eq. 3) and the observed  $\gamma$  ordering.

**Representational  $\Delta\text{RDM cosine}$  ( $L_{\text{RDM}}$ ).** The between-group geometry difference  $\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC}}$  captures which color pairs drive the CVD–HC representational deviation (§). For a candidate distortion,  $\Delta\text{RDM}_{\text{sim}}$  is obtained by applying  $\delta\theta$  to the HC mean RDM. The loss is the cosine dissimilarity between the observed and simulated 28-element upper-triangle vectors:

$$L_{\text{RDM}} = \frac{1 - \cos(\Delta\text{RDM}_{\text{sim}}, \Delta\text{RDM}_{\text{obs}})}{2} \quad (4)$$

**$hV4$  LOCO voxel-prediction ( $L_{\text{LOCO}}$ ).** The per-hue vulnerability profile  $\mathbf{v}_{\text{obs}} \in [0, 1]^8$  records where each CVD participant’s hue interpolation breaks down (§). For a candidate  $(\beta_s, \beta_c)$ , the simulated profile  $\mathbf{v}_{\text{sim}}$  is obtained by re-applying the HC-trained ridge-GCV encoding weights  $W$  (§) to the channel design matrix  $C$  shifted by the candidate distortion. The loss is the mean squared deviation between the two profiles:

$$L_{\text{LOCO}} = \frac{1}{8} \|\mathbf{v}_{\text{sim}} - \mathbf{v}_{\text{obs}}\|^2 \quad (5)$$

**Grid search.** Parameters were searched by exhaustive grid over  $(\beta_s, \beta_c) \in [-90^\circ, 90^\circ]^2$  at  $2^\circ$  resolution (8,281 cells). For R+C (Eq. 2),  $g$  was searched over  $[0, 3]$ . Per-subject loss combinations and parameter selection are described in §.

### Parameter selection

Per-subject loss combinations and model class were selected by a three-gate procedure applied to results from a 5-train/2-test HC resample ( $N = \langle 300 \rangle$ ) and a strict 7-fold HC leave-one-out (LOO) using the same atom factories.

**Gate 1: Directional precondition.** Only loss atoms where the CVD loss value exceeded the HC LOO distribution in the expected direction were admitted. Admission required signed Cohen’s  $d \geq +0.5$ ; atoms failing this gate were excluded from combination.

**Gate 2: Boundary saturation.** Among admitted combinations, those where  $\geq 50\%$  of resample solutions saturated the grid boundary were rejected as indicative of model misspecification.

**Gate 3: Held-out test-loss.** The primary criterion was the median composite test-loss ( $\bar{L}_{\text{test}}$ ) across the  $N = \langle 300 \rangle$  HC resamples. For each resample, the composite loss was evaluated at the training argmin on the two held-out HC, z-normalized relative to the training grid. Lower  $\bar{L}_{\text{test}}$  indicates generalization to unseen HC references. The secondary criterion was the test-loss IQR across resamples, measuring resample stability of the generalization estimate. A dedicated neural RDM held-out loss ( $\bar{L}_{\text{RDM}}^{\text{LOO}}$ ; Supplementary §S2) from the strict 7-fold leave-one-out isolates the neural component’s held-out performance from the behavioral weighting in the composite. Supplementary stability indices included the IQR of recovered  $(\hat{\beta}_s, \hat{\beta}_c)$  (parameter stability), the fraction of resamples recovering the modal  $45^\circ$  bin (mode share), and the strict 7-fold LOO parameter range (Supplementary §S2).

**Specificity.** The HC false-positive rate was not used as a selection criterion and is reported descriptively in § (Supplementary §S2).

**Per-subject selected loss combinations.** For sub-08 (deutan),  $\gamma_{\text{OY}} + L_{\text{RDM}}^{(\text{V2})}$  satisfied all three gates; for sub-09 (protan),  $\gamma_{\text{all}} + L_{\text{RDM}}^{(\text{V1})}$ . The LOCO loss family did not

enter the winning combination for either participant (Supplementary §S2).

### Identifiability and recovery

Four pre-specified checks were run on each production candidate before any parameter was reported. (1) Parameter recovery: synthetic CVD data at a known ground truth were forward-projected through each HC donor’s encoder with spatially correlated voxel noise, and the full fitting pipeline was re-run ( $n = 140$  per candidate); recovery was summarized by  $f_{10^\circ}$ , the fraction of refits within  $10^\circ$  of ground truth on both axes (pass  $f_{10^\circ} \geq 0.5$ ,  $|\text{bias}| < 10^\circ$ ). (2) Origin-recovery null: the same synthesis with ground truth at  $(0^\circ, 0^\circ)$  and each donor’s real JND, recording the procedure’s per-axis uncertainty floor (excluded from the test pool, as it returns no  $p$ -value). (3) HC pseudo-CVD specificity: each HC participant’s real amplitudes were substituted into the CVD slot in turn, giving a seven-member null and a one-sided rank-distance percentile. (4) Color-label permutation: trial labels of the real CVD data were shuffled  $N = 1000$  times with the HC reference held fixed, giving a one-sided  $p_{\text{perm}}$ . The three test-bearing checks across the two production candidates (six tests) were corrected together under the Benjamini–Hochberg false-discovery rate ( $\alpha = 0.05$ ); outcomes are reported in §.

### Stimulus-space filter

The subject-specific correction filter was derived as the pre-image of the fitted 2-component transform  $T(\hat{\beta}_s, \hat{\beta}_c) : \theta \mapsto \theta + \delta\theta(\theta; \hat{\beta}_s, \hat{\beta}_c)$ . For each of the 8 displayed hues  $\theta_k$ , we solved for the input angle  $\tilde{\theta}_k$  such that  $T(\tilde{\theta}_k) = \theta_k$ , yielding a per-hue correction  $\delta\theta_k^{\text{filt}} = \tilde{\theta}_k - \theta_k$ . A CVD observer viewing  $\tilde{\theta}_k$  thus receives, under the model, the same cortical hue representation as a healthy control observer viewing  $\theta_k$ .

Pre-images were computed by numerical inversion of the forward map (Brent’s-method root-finding) to a convergence tolerance of  $< 0.001^\circ$ .

### Behavioral tasks

***Just-noticeable difference (JND)..*** Color discrimination thresholds were estimated for eight hue pairs using two-interleaved 1-up/1-down adaptive staircases,

each terminating after eight reversals (Levitt, 1971). Pairs spanned three categories: (i) three pairs hypothesized as universally difficult for both CVD types (yellow–purple, blue–purple, red–orange), (ii) two pairs probing the deutan confusion axis (orange–yellow, yellow–green), (iii) two pairs probing the protan confusion axis (cyan–magenta, green–blue), and (iv) one control pair (red–cyan). For each pair, the JND ratio  $\gamma = \gamma_{\text{CVD}}/\bar{\gamma}_{\text{HC}}$  was computed, where  $\bar{\gamma}_{\text{HC}}$  is the HC mean threshold. Ratios  $\gamma > 1$  indicate elevated discrimination difficulty relative to HC. These ratios enter the fitting loss as described in §.

**Filter evaluation (sub-08).** The deutan participant (sub-08) completed a second scanning session evaluating two filters against the unfiltered Session-1 baseline; the protan participant was not collected. The *personalized* filter was the per-subject 2-component stimulus-space pre-image with parameters frozen from the main analysis ( $\hat{\beta}_s = 6^\circ$ ,  $\hat{\beta}_c = -42^\circ$ ), rendered directly in PsychoPy. The *deployed* comparator was the macOS accessibility Color Filter (System Settings > Accessibility > Display, build 26.5.1) at an intensity the participant self-tuned before scanning; it is the unmodified shipping product, an operating-system-level post-display transform. We compared against the deployed accessibility filter rather than a re-implemented retinal-simulation transform so that the comparator is the correction an end user actually encounters. Because the filter parameters were frozen before this session, it is out-of-sample with respect to fitting. Eight fMRI runs (four per filter) were acquired in ABBA-counterbalanced order. On the second-session data we recomputed, within each condition, LOCO forward-tuning correlation, LORO classification, SRM Procrustes disparity into the HC shared space, and the representational dissimilarity matrices. Each condition’s neural index was compared to the HC distribution as a standardized single-case effect size (Cohen’s  $d$  against the HC mean and SD, with the HC runs subsampled to four to match the per-condition run count), reported in place of an inferential  $p$  because the grand-mean permutation null is biased for this design; robustness was assessed by leave-one-run separation and by re-extracting both conditions on the no-filter baseline voxel set. Behavioral testing was repeated under



both filters: JND staircases for the eight hue pairs and an eight-alternative RSVP color-identification task (64 trials), each compared to the unfiltered Session-1 baseline and to the HC distribution. This evaluation is a single deutan case (four runs per condition); the deployed and personalized filters use different rendering pipelines, so differences between them are interpreted as movement toward or away from the HC reference, not as evidence for a corrective mechanism, and the interleaved, cross-session behavioral comparison is descriptive.

## Reproducibility

All analyzes were conducted in Python 3.10 (numpy 1.24.3, scipy 1.11.3, scikit-learn 1.3.0, BrainIAK 0.11). Random seeds were fixed (`seed=42`) for all permutation tests and bootstrap resampling. Analysis code and preprocessing scripts are available at [https://github.com/Transconnectome/colorBlind\\_analysis](https://github.com/Transconnectome/colorBlind_analysis). Anonymized preprocessed data are available on reasonable request to the corresponding author.

## Results

### Color discrimination is preserved across all ROIs in both CVD cases

A corrective stimulus-space filter is valid only if the visual cortex of a CVD individual retains categorical discrimination among the target hues. We verified this precondition using leave-one-run-out (LORO) cross-validation of the forward encoding model in SRM-aligned space.

Both CVD participants exceeded the 0.125 exact-accuracy chance level at every ROI (Figure 4A). A cross-subject generalization test of the same forward encoding model (HC-to-HC vs. HC-to-CVD pair-wise cross-decoding, all ROIs pooled) showed no HC-CVD difference (Mann-Whitney  $U$  test, 21 HC-to-HC vs. 14 HC-to-CVD pairs,  $p = 0.668$ , rank-biserial effect size and exact test statistic in Supplementary ). The same null held at the within-ROI level at hV4 (Crawford & Howell  $t$ -test, both CVD subjects within the HC range,  $p = 0.142$ ). The 8 scanner stimuli are therefore individually

distinguishable at the cortical level in both CVD participants, in line with prior work on CVD above-threshold identification (Boehm et al., 2014; Bosten, 2019).

### Color interpolation is impaired in both CVD cases

We next tested whether continuous hue interpolation fails, which indicated structural impairment in cortical representation, using leave-one-color-out (LOCO) decoding of the forward encoding model. Among the four retinotopic ROIs, only hV4 supported above-chance hue interpolation in HC (adjacent accuracy  $0.47 \pm 0.05$ , mean  $\pm$  SEM over  $n = 6$  HC, sub-07 excluded at hV4 for insufficient voxels;  $p = 0.044$  under  $8! = 40,320$  exact label permutations). V1–V3 did not exceed chance despite preserving cross-run discrimination. This is consistent with hV4 as the primary interpolation gate, replicating Brouwer and Heeger (2009).

Both CVD participants fell at or below the adjacent-accuracy chance level at hV4 (Figure 4B). Sub-08 reached adjacent accuracy 0.25 (Crawford & Howell  $t = -1.58$ ,  $p = 0.082$ , n.s.,  $d_{cc} = -1.71$ ). Sub-09 reached 0.13 ( $t = -2.48$ ,  $p = 0.024$ ,  $d_{cc} = -2.68$ ). The impairment was not uniform across hues. Both participants showed near-zero adjacent accuracy at hV4 for the S-cone intermediate colors blue, purple, and magenta (adjacent accuracy = 0 for both participants at all three hues; Figure 4C). Per-hue Crawford & Howell tests (one-tailed) reached significance only for blue (both participants  $d = 2.20$ ,  $p = 0.042$ ); purple ( $d = 1.02$ ,  $p = 0.19$ ) and magenta ( $d = 1.89$ ,  $p = 0.064$ ) did not survive the single-case test against the  $n = 6$  HC distribution. These per-hue accuracies form each participant’s hue vulnerability profile  $\mathbf{v} \in [0, 1]^8$ .

### Color geometry is distorted at a distinct ROI in each CVD case

To characterize the geometric basis of the LOCO impairment, we compared the pairwise representational structure of CVD and HC participants in a shared SRM-aligned space. For each CVD participant we computed disparity and the per-pair RDM deviation from the HC mean ( $\Delta\text{RDM} = \text{RDM}_{\text{CVD}} - \overline{\text{RDM}}_{\text{HC}}$ ).

The ROI of elevated disparity differs between the two CVD participants: V2 in sub-08 and V1 in sub-09 (Figure 5B). Sub-08 showed significantly elevated disparity

specifically at V2 (Crawford & Howell  $p = 0.040$ ), with no significant elevation at V1, V3, or hV4. Sub-09 showed significantly elevated disparity specifically at V1 ( $p = 0.007$ ), with no significant elevation at V2, V3, or hV4. Full test statistics and effect sizes ( $t$ ,  $d_{cc}$ ) for these Crawford–Howell comparisons are reported in Supplementary .

The  $\Delta$ RDM heatmaps (Figure 5A) reveal subject-specific distortion structures. Sub-08 shows elevated distances among S-cone intermediate pairs at V2, and sub-09 shows a broader reorganization of pairwise distances at V1. Disparity tests the absolute amplitude of pairwise distances against the HC leave-one-out distribution, whereas the pairwise  $\Delta$ RDM shows the directional pattern of those deviations; the significant V1/V2 disparity is independent evidence that the cortical color representation is geometrically distorted.

### **The retinal-plus-gain model is structurally insufficient**

The R+C model is structurally insufficient for both CVD participants. Fitting this model to each CVD participant yielded ceiling saturation in both cases. For sub-08,  $\langle 100\% \rangle$  of resample solutions saturated the grid boundary ( $g = 3.0$ ). For sub-09, saturation remained high ( $\langle 41\% \rangle$ ,  $g = 2.95$ ). Boundary saturation at these rates indicates model misspecification, not a well-constrained optimum (Wilson & Collins, 2019).

A gain  $g > 2$  implies that cortical processing reverses the direction of the retinal shift past the undistorted hue (Section ). Both participants scored below the Ishihara criterion (sub-08: 5/14 plates correct; sub-09: 7/14), confirming residual CVD under standard viewing conditions. A fitted  $g > 2$  is therefore internally inconsistent with confirmed CVD status: the model claims overcompensation where the participants demonstrably show impairment. Therefore, the saturated boundary is a model failure, not a valid parameter estimate.

The structural cause is a degree-of-freedom deficit. The function  $\delta\theta = (2 - g) \cdot \delta\theta_{\text{Mach}}$  displaces hues only along the fixed confusion axis. No value of  $g$  can account for distortions in directions orthogonal to that axis. R+C is therefore

excluded as a filter candidate.

### A common cortical model fits both CVD cases

The 2-component cortical model fit both participants using the same functional form. The deutan and protan cases differ in  $(\hat{\beta}_s, \hat{\beta}_c)$ , not in model structure. We fit the two-component parameters  $(\beta_s, \beta_c)$  for each participant and inverted the fitted transform to a stimulus-space filter.

For sub-08 (deutan,  $\theta_{\text{conf}} = 150^\circ$ ), the selected loss combination  $\gamma_{\text{OY}} + L_{\text{RDM}}^{(V2)}$  yielded  $(\hat{\beta}_s, \hat{\beta}_c) = (6^\circ, -42^\circ)$ . The held-out test-loss was  $\bar{L}_{\text{test}} = -2.36$  (IQR = 2.15), lower than the alternative candidate ( $\bar{L}_{\text{test}} = -1.14$ ; Supplementary §S2). The HC-resample parameter IQR was  $(8^\circ, 2^\circ)$ . The strict 7-fold LOO  $\hat{\beta}_c$  range was  $[-46^\circ, -38^\circ]$ ; all values are negative, confirming a stable sign across held-out reference sets.

For sub-09 (protan,  $\theta_{\text{conf}} = 16^\circ$ ), the selected loss combination  $\gamma_{\text{all}} + L_{\text{RDM}}^{(V1)}$  yielded  $(\hat{\beta}_s, \hat{\beta}_c) = (2^\circ, +24^\circ)$ . The held-out test-loss was  $\bar{L}_{\text{test}} = -1.54$  (IQR = 1.42). The HC-resample parameter IQR was  $(0^\circ, 0^\circ)$ , and  $\langle 87.7\% \rangle$  of resamples recovered the same  $45^\circ$  bin. The strict 7-fold LOO IQR was  $(0^\circ, 0^\circ)$ . However, this stability is specific to the PCA-basis loss: the argmin shifts under the alternative SRM-basis reduction, so the recovered position is metric-dependent.

The RDM atom in each subject’s winning combination was consistent with the ROI showing significant geometric disparity: V2 for sub-08 ( $p = 0.040$ ) and V1 for sub-09 ( $p = 0.007$ , Section ). The two criteria were independent: the loss combination was selected by held-out test-loss, and the significant disparity ROI was identified by Crawford–Howell tests on disparity amplitude.

On held-out folds, both fitted distortions reproduced the observed CVD geometry, beating the no-correction baseline on all 7 folds and ranking in the top 5–8% of grid combinations. Relative to the HC noise ceiling — the best fit achievable given HC data noise, estimated from split-half HC amplitude reliability (Lage-Castellanos, Valente, Formisano & De Martino, 2019) and conservative because CVD amplitude noise is

excluded — the fits reached 52% (sub-08) and 67% (sub-09) of the way from baseline to ceiling (held-out losses and ceiling values in Supplementary §S2).

### Neural data identifies what behavior cannot

Fitting behavioral ( $\gamma$ ) and neural (RDM) loss atoms independently reveals three benefits of utilizing the neural component.

First, the neural term captures a protan cortical direction that the behavioral loss cannot resolve. Sub-09’s behavioral-only fit yielded  $\hat{\beta}_c \approx +4^\circ$  and did not beat the no-correction baseline ( $\Delta L = +0.01$ , 4/7 folds). The RDM term captures a signal that the behavioral loss cannot detect for this participant.

Second, the neural term stabilizes the argmin for sub-08. The behavioral-only and combined fits share the same argmin ( $6^\circ, -42^\circ$ ), confirming that the behavioral atom independently locates the deutan direction; the neural-only fit independently corroborates this direction with a meaningful estimate ( $\hat{\beta}_c = -26^\circ$ ; Supplementary §S2). Adding the RDM atom reduced the boundary saturation rate from 23% to 9.3%, sharpening the argmin without shifting it.

Third, the neural term enhances parameter stability by reducing its IQR in both participants. For sub-08, adding the neural term reduced the HC-resample IQR from  $(18^\circ, 6^\circ)$  to  $(8^\circ, 2^\circ)$  (PCA) and  $(10^\circ, 4^\circ)$  (SRM). For sub-09, the IQR narrowed from  $(6^\circ, 4^\circ)$  to  $(0^\circ, 0^\circ)$  (PCA, deterministic) and  $(0^\circ, 2^\circ)$  (SRM).

### Identifiability: mechanism class recoverable, per-axis magnitude not

These fits recover the distortion’s mechanism class and the direction of its dominant axis, but not its precise per-axis magnitudes. Parameter recovery fails across all pre-specified identifiability checks (0/6 significant after FDR correction, BH  $\alpha = 0.05$ ; Supplementary §S3). The voxel-level parameter-recovery test yielded  $f_{10^\circ} < 0.30$  for both production candidates. The non-dominant axes ( $|\hat{\beta}_s| \leq 6^\circ$  for both participants) fall below the effective parameter uncertainty of  $\sim 20^\circ$ – $25^\circ$  and are not recoverable (Supplementary §S3). The dominant axis ( $|\hat{\beta}_c| = 42^\circ$  for sub-08) exceeds this uncertainty and is partially recoverable: when CVD responses are simulated from a

known distortion and re-fitted, the recovered  $\hat{\beta}_c$  deviates from the true value by  $4.7^\circ$  on average (Supplementary §S3).

Consistent with the partial recoverability of the dominant axis, the sign of  $\hat{\beta}_c$  — the distortion direction — is stable across held-out reference sets (Section ). Sub-08 consistently returns  $\hat{\beta}_c < 0$  across all LOO folds and resamples. Sub-09 consistently returns  $\hat{\beta}_c > 0$  under the PCA-basis loss; sign stability under the SRM-basis loss was not verified (Supplementary §S3). The distortion direction is recoverable even when its magnitude is not.

The loss landscape shows where this magnitude indeterminacy lies. Although each participant’s selected fit is the global minimum on the full data (Figure 6), the per-resample argmins do not all concentrate there. For sub-09, 87.7% recur at the selected fit, but a secondary basin near  $(32^\circ, 0^\circ)$  marks a metric-dependent alternative. For sub-08, they disperse along the  $\beta_c$  axis, leaving its magnitude loosely constrained.

### Per-subject stimulus-space filter

The per-subject filter  $\delta\theta_k^{\text{filt}} = \tilde{\theta}_k - \theta_k$  is the pre-image of the fitted 2-component transform (Section ).

For sub-08 (deutan,  $\hat{\beta}_s = 6^\circ$ ,  $\hat{\beta}_c = -42^\circ$ ,  $\theta_{\text{conf}} = 150^\circ$ ), the dominant  $\hat{\beta}_c$  term produces a large-amplitude correction that varies asymmetrically across hues (mean  $|\delta\theta| = 26.3^\circ$ ; Figure 7).

For sub-09 (protan,  $\hat{\beta}_s = 2^\circ$ ,  $\hat{\beta}_c = +24^\circ$ ,  $\theta_{\text{conf}} = 16^\circ$ ), the smaller  $\hat{\beta}_c$  produces a more targeted correction concentrated near the protan confusion axis (mean  $|\delta\theta| = 16.2^\circ$ ; Figure 7).

The two fits differ in the magnitude and direction of the dominant confusion-axis term ( $\hat{\beta}_c = -42^\circ$  for sub-08 vs.  $+24^\circ$  for sub-09).

### Behavioral and neural filter evaluation (sub-08)

We evaluated the personalized filter in a second session in the deutan participant (sub-08; sub-09 was not collected). On the primary neural index — LOCO forward-tuning correlation to the canonical hue map — the personalized filter was more

healthy-control-like than the deployed macOS accessibility filter in all four ROIs (Fig. 8A; personalized  $\rho$ : V1 +0.21, V2 +0.10, V3 +0.05, V4 +0.18; deployed  $\rho$ : V1 -0.32, V2 -0.19, V3 -0.32, V4 -0.39). The contrast was robust: the personalized condition remained more HC-like after dropping any single run (leave-one-run separated in every ROI) and after re-extracting both conditions on the baseline voxel set (personalized-deployed  $\Delta\rho = 0.37\text{--}0.57$ ). The deployed filter drove forward-tuning strongly negative, distorting rather than restoring the canonical hue map. The effect holds across ROIs rather than at V1 alone: at the baseline voxel set the personalized filter sat at the HC level at V2 ( $d = +0.04$ ) and within one HC standard deviation at V3 and V4, while V1's larger native effect ( $d = +0.97$ ) attenuated under voxel matching ( $d = -0.82$ ), indicating it was partly driven by occipital coverage.

The filters were not separable on every readout. LORO classification accuracy was near-equal across the deployed, personalized, and HC conditions ( $\approx 0.69\text{--}0.72$  at V1), so both filters preserved color decodability while only the personalized filter restored the forward-tuning geometry. SRM disparity favored the personalized over the deployed filter in all ROIs, but the no-filter baseline had the lowest disparity of the three (surviving run-count matching), and the shared-space RDM agreed: neither filter was more HC-like than no filter by representational geometry (Fig. 8C,D). The personalized filter thus restores forward-tuning to HC levels but does not normalize the color geometry. That geometry remains unlike HC's and more compressed.

Behaviorally, both filters worked: the baseline deutan deficit (JND mean  $|z| = 2.24$ ; the orange-yellow, yellow-green, and yellow-purple pairs all exceeding the HC empirical maximum) was eliminated by both the deployed filter ( $|z| = 0.85$ ) and the personalized filter ( $|z| = 0.78$ ), and RSVP identification accuracy rose from 0.81 at baseline to 0.97 under both, abolishing the yellow-green and purple-magenta confusions. The two filters could not be ranked behaviorally (Wilcoxon  $p = 0.84$ ) and failed on different pairs (the personalized filter over-corrected orange-yellow below the HC range); the single-subject, single-run, cross-session design with a shared-pipeline confound precludes a behavioral ranking. Both a well-configured deployed filter and the

personalized filter thus restore sub-08’s discrimination to near-control levels, and the personalized filter’s distinctive advantage is confined to the pipeline-robust neural forward-tuning contrast.

## Discussion

We asked whether the cortical color representation of a color-vision-deficient (CVD) individual is structured enough to ground a personalized corrective filter. Two CVD participants, one deutan and one protan, retained categorical discrimination of all eight color hues at every retinotopic ROI. Yet both selectively lost continuous-hue interpolation, falling at or below the adjacent-accuracy chance level. This deficit was not diffuse signal loss but a structured geometric distortion. We modeled it with two cortical components: a rotation along the subtype confusion axis — the classical direction of CVD color confusion — and a distortion along the S-cone axis, where the interpolation deficit was concentrated. Fit independently per participant from behavioral discrimination thresholds and cortical representational geometry, this model recovered an exact stimulus-space pre-image for all eight hues in both cases. That pre-image is the per-person correction filter, built from that individual’s own cortical color geometry. We show, in the deutan case, that such a filter pulls the cortical hue geometry back toward the healthy-control reference; behavioral discrimination was restored, but not beyond a deployed accessibility filter.

### A geometric distortion of cortical color representation

In both CVD participants, the deficit appeared as a structured distortion of cortical color geometry rather than a uniform attenuation of the chromatic signal. Representational geometry treats the pairwise distance structure among stimuli as an encoded relationship in each cortical region (Kriegeskorte, Mur & Bandettini, 2008; Kriegeskorte & Douglas, 2018). Two complementary measurements expose a structural distortion, each at the stage where its signal exists. The representational dissimilarity matrix (RDM) shows *which* pairwise distances deviate from the healthy-control (HC) geometry: disparity was significantly elevated at V2 in the deutan participant



(Crawford & Howell  $p = 0.040$ ) and at V1 in the protan participant ( $p = 0.007$ ). The two participants showed this elevated disparity at different areas (V2 and V1), not a common one. Leave-one-color-out decoding shows *where* the continuous-hue manifold breaks down. Only hV4 supported above-chance interpolation in HC ( $0.47 \pm 0.05$ ,  $p = 0.044$ ), replicating Brouwer and Heeger (2009). There, the protan participant fell significantly below HC (0.13,  $p = 0.024$ ; exploratory single-case), and the deutan participant showed a marginal reduction in the same direction (0.25,  $p = 0.082$ ). Both sat at zero adjacent accuracy, with a significant deficit at the S-cone intermediate blue ( $d = 2.20$ ,  $p = 0.042$ ). This account aligns with prior characterizations of CVD as a multidimensional deformation of color space rather than a uniform attenuation (Boehm et al., 2014; Emery et al., 2021).

### A neurally grounded, individualizable correction filter

We built each correction filter by inverting that participant’s fitted two-component model to its stimulus-space pre-image (Methods §) — the displayed hue that, after the cortex distorts it, is perceived as the intended color. Across the eight hues this filter applied a mean rotation of  $|\delta\theta| = 26.3^\circ$  for the deutan participant and  $16.2^\circ$  for the protan participant.

Fitting the behavioral and neural loss atoms independently revealed three benefits of grounding the correction in cortical geometry. First, the neural term recovered a confusion-axis direction the behavioral loss could not resolve: the protan behavioral-only fit did not beat the no-correction baseline ( $\Delta L = +0.01$ , 4 of 7 folds), whereas the RDM term resolved a clear rotation. Second, it stabilized an already-informative solution: for the deutan participant the behavioral-only and combined fits shared the same argmin ( $6^\circ, -42^\circ$ ), and adding the neural term sharpened it without shifting it, lowering the boundary-saturation rate from 23% to 9.3%. Third, it tightened parameter stability in both participants, narrowing the resample interquartile range of the recovered parameters. The neural-only fit for the deutan participant independently returned the same direction ( $\hat{\beta}_c = -26^\circ$ ), corroborating the

behavioral estimate. A correction grounded in individual cortical geometry therefore differs in substance from a purely behavioral one.

The resulting filter is per-person and neurally grounded — the architectural advance over a population-average correction. The two fitted filters differ by subtype in the *direction* of the dominant confusion-axis term ( $\hat{\beta}_c = -42^\circ$  for the deutan participant versus  $+24^\circ$  for the protan participant). Per-axis magnitude is not identifiable (below) and is not the basis of the contrast. The architecture is individualizable by construction: each filter is built from one participant’s measured cortical color geometry rather than from a fixed subtype-average spectral shift. Because  $\theta_{\text{conf}}$  is subtype-defined and the sample holds one deutan and one protan, this contrast is between-subtype. What makes each correction individual is that its distortion is read from that participant’s own cortical color geometry, not assigned from their diagnostic subtype. Across held-out reference sets, the two-component mechanism class and the sign of the dominant confusion-axis term are stable — the level the filter relies on. Their per-axis magnitudes are not separately identifiable (0 of 6 recovery checks survive FDR correction), so the between-subtype contrast rests on these robust directions rather than on absolute parameter values.

A population-average transform does not substitute for this per-person inverse, for a reason that holds independent of any fit. The Machado class — a fixed spectral shift along the single confusion axis, even when scaled by a cortical gain (Machado et al., 2009) — stays confined to that one axis and therefore does not represent distortion in orthogonal directions. For the protan participant it even collapses three distinct displayed hues onto a single pre-image near  $127^\circ$ , leaving no exact pre-image where that participant is most vulnerable (Appendix ). Fitting the class confirmed the mismatch empirically — its gain saturated near  $g \approx 3$ , implausibly implying overcompensation in observers who remain measurably impaired — but the structural limit alone rules it out as an exact correction (Section ).

### Filter evaluation in the deutan case

We tested the personalized filter against a deployed macOS accessibility filter in a second session in sub-08 (the protan was not collected). On cortical forward-tuning — the leave-one-color-out geometry that was selectively impaired — the personalized filter moved sub-08 toward the healthy-control reference in all four ROIs and outperformed the deployed filter, which instead distorted the forward map. Discrimination (LORO) was preserved under both filters, mirroring the discrimination–interpolation dissociation that defines the deficit: the population-average filter restores categorical decodability but not the continuous geometry, whereas the personalized filter restores both. Behaviorally, both filters improved discrimination from the baseline deficit to near-control levels — both worked — and the two could not be ranked against each other. On raw representational geometry (SRM), by contrast, neither filter was more healthy-control-like than the unfiltered baseline, bounding the geometry claim to the forward-tuning measure. The personalized filter’s distinctive advantage is thus confined to the pipeline-robust forward-tuning contrast, in a single deutan case; a behavioral advantage and generalization beyond this individual remain open.

### Limitations

Four considerations bound the proof-of-concept scope of these findings. First, the CVD sample is  $N = 2$ . Single-case statistics (Crawford & Howell, 1998) permit individual-level inference, but population-level claims about deutan or protan subtypes require replication with more participants. As established above, the sign of  $\hat{\beta}_c$  is the recoverable quantity. Whether the between-subtype  $\hat{\beta}_c$  difference ( $-42^\circ$  vs.  $+24^\circ$ ) is subtype-level or idiosyncratic awaits power-adequate replication. Establishing per-person rather than subtype-average correction specifically requires testing several individuals *within* a single subtype — the decisive follow-up experiment. Second, the stimulus set was confined to a single isoluminant, iso-chroma locus ( $L^* = 75$ , chroma = 40). The filter is therefore derived and, in Phase 3, validated only along the hue circle at fixed lightness and saturation. It does not constrain how the correction generalizes to

luminance- or saturation-varying stimuli or to naturalistic viewing — gaps that denser stimulus sampling across these axes would close. Third, the HC reference cohort is small (seven controls), so the HC  $\|\hat{\beta}\|$  leave-one-out distribution we report is a magnitude anchor for the CVD estimates, not a hypothesis test. A larger HC reference cohort would be required to arbitrate population-level distortion patterns. Fourth, we did not compute parameter-level bootstrap confidence intervals on  $\hat{\beta}_s$  and  $\hat{\beta}_c$ , so the per-axis estimates are point values whose interval precision awaits the additional runs per participant that bootstrap resampling would require.

## Conclusion

This work is a proof-of-concept that each individual’s cortical color geometry, measured with fMRI, can ground a stimulus-space correction obtained by inverting a fitted model of that individual’s representational distortion. It illustrates, in two cases, that a color-vision deficiency can present not as a uniform retinal loss to be undone by a fixed subtype-average spectral shift, but as an individually patterned, multidimensional distortion of the cortical color representation — one structured enough to invert. The architecture that follows — measure an individual’s cortical geometry, fit its distortion, and invert it to a per-person correction — is distinct by construction from population-average retinal approaches. A fixed subtype-average spectral shift does not represent a per-person distortion, whereas a per-person inverse does, for the mechanism class and up to the per-axis magnitude uncertainty quantified above. In the deutan case, this cortical-level correction moved forward-tuning toward the healthy-control reference beyond a deployed filter; a behavioral advantage and broader validation remain to be established. A correction spanning hue, lightness, and chroma jointly — fitted from denser stimulus sampling, more runs per participant, and a larger HC reference pool — is the path toward a deployable personalized filter. More broadly, treating a sensory deficit as an invertible distortion of its own cortical representation — characterized and corrected one individual at a time — offers a template for personalized, brain-based perceptual correction beyond color vision.

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## Appendix A: Alternative Model Comparison

The main text adopts a 2-component cortical model (Section ). Here we report two cone-shift baselines as reference comparisons:

- **Machado (1-DOF)** — a single retinal cone-spectral shift  $\Delta\lambda$  Machado et al., 2009.
- **Retinal + Cortical (R+C, 2-DOF)** — retinal shift  $\Delta\lambda$  plus a cortical compensation gain  $g$  Tregillus et al., 2021.

### A.1 Fit results (LOCO Spearman $\rho$ argmax, historical reference)

Under the LOCO- $\rho$  argmax criterion used in earlier drafts, the Machado model fit Sub-09 at  $\Delta\lambda = 13.5$  nm with  $\rho_{V4} = 0.76$  and Sub-08 at  $\rho_{V4} = 0.62$  (lower fit quality); the R+C model did not improve on either single-component model in either subject. These fits are reported as historical reference for the main-text 2-component fit (Section ); the LOCO- $\rho$  argmax criterion used here need not agree with the per-subject loss selected in the main text (§).

***R+C decomposition as a diagnostic.*** Although the R+C 2-stage model is not adopted as the filter form (see Per-subject stimulus-space filter), the R+C parameter fit at hV4 is reported here as an exploratory per-subject diagnostic. For sub-08, the R+C decomposition returns  $\Delta\lambda = 2.5$  nm with  $g = -2.25$ ; for sub-09 it returns  $\Delta\lambda = 19.5$  nm with  $g = -1.10$ . These values are reported for completeness only and carry no etiological interpretation. The R+C neural loss is weakly identified (near-degenerate) along the  $\Delta\lambda$ - $g$  trade-off, and the  $g$  estimate in particular is unstable across fitting variants, reversing sign under alternative criteria (the historical LOCO- $\rho$  argmax fit returns  $g = +2.25$  for sub-08, versus  $g = -2.25$  here; Supplementary Table 1); the  $(\Delta\lambda, g)$  pair therefore does not constitute a reliable point estimate and supports no retinal-versus-cortical attribution. We retain it only as a descriptive companion to the 2-component fit, not as an independent test.

## A.2 Inverse-mapping (pre-image) behavior

The angular form of the 2-component model is bijective by construction: a stimulus-space pre-image  $\theta + \delta\theta$  exists for any target hue. The 1-DOF Machado fit for sub-09, by contrast, collapses three hues (green  $135^\circ$ , cyan  $180^\circ$ , blue  $225^\circ$ ) onto a single pre-image angle ( $\sim 127^\circ$ ), which mathematically prevents exact inversion at those positions. This arc-compression failure is a structural property of the 1-DOF retinal-shift model and is independent of the fitting criterion; it disqualifies the Machado baseline for the stimulus-space filter task of Section .

## A.3 R+C model

The R+C model is retained here as a 2-DOF baseline. Because the cortical gain term  $g$  acts as a uniform amplitude scaling without generating the angular distortion captured by the 2-component model's  $\beta_c \cos(h - \theta_{\text{conf}})$  term, R+C does not match the observed hV4 LOCO vulnerability pattern beyond what the retinal  $\Delta\lambda$  term alone provides; this conclusion is consistent across loss criteria.

## Supplementary Methods

### S1. Confound Regression and Temporal Filtering

Within-run head motion was estimated via rigid-body realignment. No temporal filtering or confound regression was applied; slow drift was modeled via linear per-run regressors in the general linear model.

### S2. Quality Control

Registration quality was evaluated by computing the intersection between the Wang Atlas ROIs (V1, V2, V3, hV4) and the BOLD brain mask in MNI space. Across 10 subjects, mean ROI coverage was 84.3% (SD = 21.7%), and GLM valid ratio (percentage of ROI voxels with reliable stimulus-evoked responses) was 99.6%. We evaluated all subjects' data suitable for downstream analysis, though sub-07 showed reduced coverage (30.8%) due to individual anatomical variability.

### S5. Mean Activation Analysis

To verify that the shared-space geometric differences are not a signal-quality confound, we compared activation-level metrics between groups before SRM alignment. Mean activation magnitude (mean  $|\beta|$ ), signal-to-noise ratio, color modulation depth, run-to-run reliability, and spatial variance did not differ between HC and CVD across V1, V2, V3, and hV4 (Welch's  $t$ -tests, all  $p > 0.09$ ), and the two CVD cases fell within the HC range on every metric (Crawford–Howell, all  $p > 0.09$ ). Voxel-level color selectivity was strong in both groups (one-way ANOVA across voxels,  $F > 4$ ,  $p < 0.001$  in V1–V3). Critically, none of the activation metrics correlated with SRM disparity across participants (Pearson  $r$ , 20 tests, all  $p > 0.13$ ). The HC–CVD difference is therefore one of representational pattern, not signal amplitude: the cortical color response is equally strong in CVD but geometrically reorganized.

## S6. Statistical Analysis

All tests used a significance threshold of  $\alpha = 0.05$  with False Discovery Rate correction (Benjamini & Hochberg, 1995);  $q = 0.05$  applied to RDM color-pair comparisons (28 pairs per participant). Individual CVD participants were compared to the HC distribution using Crawford and Howell (1998) modified t-tests:

$$t^* = \frac{x_{\text{CVD}} - \bar{x}_{\text{HC}}}{\text{SD}_{\text{HC}} \times \sqrt{(n+1)/n}}, \quad df = n - 1 = 6 \quad (6)$$

with one-tailed p-values testing the directional hypothesis that CVD disparity exceeds HC. Effect sizes were quantified via Hedges'  $g$ . Group-level permutation tests (10,000 iterations) were used for HC-CVD comparisons by permuting subject labels; color-level permutation tests (1,000 iterations) were used for LOCO/LORO validation by permuting color labels within subjects.

## S7. Dimensionality Selection (K-Selection)

The reduced dimension  $k$  for SRM was determined via leave-one-subject-out (LOSO) cross-validation based on within-subject representational dissimilarity matrix (RDM) reliability (Cunningham & Yu, 2014). For each candidate  $k$  (range: 1–8) and fold, reliability was quantified as the Spearman correlation between RDMs computed from even runs (2,4,6) versus odd runs (1,3,5) of the held-out participant in SRM space. We selected  $k$  maximizing mean reliability across seven folds. The selected values and their associated cross-subject RDM Spearman correlations were:

- V1:  $k = 4$  (cross-subject RDM correlation = 0.60)
- V2:  $k = 4$  (cross-subject RDM correlation = 0.57)
- V3:  $k = 3$  (cross-subject RDM correlation = 0.55)
- hV4:  $k = 3$  (cross-subject RDM correlation = 0.32)

### S8. Generalized Cross-Validation for Ridge Regression

The regularization parameter  $\alpha$  for ridge regression was selected via Generalized Cross-Validation (GCV; Golub et al., 1979), implemented as `sklearn.linear_model.RidgeCV` with `gcv_mode='auto'` (Pedregosa et al., 2011). GCV provides a computationally efficient approximation to leave-one-out cross-validation for selecting  $\alpha$ . The optimal  $\alpha$  minimizes the GCV score:

$$\text{GCV}(\alpha) = \frac{1}{n} \|y - \hat{y}\|_2^2 / \left(1 - \frac{\text{tr}(H)}{n}\right)^2 \quad (7)$$

where  $X$  denotes the design matrix,  $y$  is the observed response vector,  $\hat{y}$  is the ridge-predicted response,  $H = X(X^T X + \alpha I)^{-1} X^T$  is the hat matrix, and  $n$  is the number of training samples.

### S9. Leave-One-Out Consistent Disparity Estimation

To quantify HC-CVD disparity with unbiased references, we used a leave-one-out (LOO) consistent procedure: for each fold  $i$  (leaving out HC participant  $i$ ), a reference pattern  $R_{-i}$  was computed as the mean of the remaining six HC participants' aligned data. Both the held-out  $\text{HC}_i$  and both CVD participants were compared against this same reference  $R_{-i}$ , yielding disparity scores  $d(\text{HC}_i, R_{-i})$  and  $d(\text{CVD}_j, R_{-i})$ . This ensures HC and CVD disparities are measured against identical references within each fold. Individual CVD disparity significance was assessed via Crawford and Howell (1998) modified t-tests comparing each CVD participant's mean disparity (averaged across 7 folds) to the HC distribution ( $n = 7$ ,  $df = 6$ ).

### S10. Cross-Validation Procedures

#### Leave-One-Run-Out (LORO)

LORO evaluated whether individual colors could be distinguished from voxel response amplitudes, testing the consistency of neural color encoding across runs. For each of six runs, the corresponding response matrices ( $B$ ,  $C$ ) were held out as the test

set,  $W$  was trained on the remaining five runs. Both color decoding and voxel response prediction were evaluated on the held-out run.

### **Leave-One-Color-Out (LOCO)**

LOCO tested whether geometric relationships of circular hue space were preserved in neural representation by evaluating interpolation to held-out colors. For each color, all six run response matrices of the target color were held out.  $W$  was trained on the remaining seven colors, and the model predicted either the held-out color from voxel responses or voxel responses from the held-out color’s channel outputs.

### **Leave-One-Subject-Out (LOSO)**

LOSO evaluated zero-shot generalization to held-out participants. As LOSO requires cross-subject prediction, SRM-transformed data were used to align voxel spaces across participants. For each HC participant,  $W$  was trained on the remaining HC participants’ shared-space data and tested on the held-out participant. This assessed whether the HC group-derived common space supported generalization to novel individuals.

## **S11. Evaluation Metrics**

### **Color Decoding Metrics**

Color decoding was evaluated using (1) mean absolute error (MAE) between predicted and true hue angles (chance =  $90^\circ$ , expected from a uniform random predictor on a circle), and (2) categorization accuracy with  $\pm 22.5^\circ$  bins around each of the eight colors (chance =  $12.5\% = 1/8$ ).

### **Voxel Prediction Metrics**

Voxel response prediction was evaluated using Pearson correlation between predicted and observed response vectors, computed across all voxels and colors in the test set.

## Statistical Significance

Statistical significance of group comparisons (HC vs CVD) and individual CVD analyzes was assessed using nonparametric permutation tests (10,000 iterations; Nichols & Holmes, 2002).

## S12. Shared Response Model Residuals

Each participant’s residuals in the SRM model reflect the extent to which their data are not captured by the common structure, including inherent geometric differences. The orthogonal constraint ( $W_i^T W_i = I_k$ ) helps preserve geometric relationships between colors—distances and angles—in the common space.

Additionally, global between-subject RDM similarity was quantified using Spearman correlation across all 28 color pairs.

## S13. Image Orientation Initialization

Image orientation was first initialized according to the scanner-defined obliquity information in the NIFTI header. The mutual-information cost function was then optimized to maximize the statistical dependence between T1w and BOLD intensity distribution within the overlapping region.

## S14. Procrustes Alignment Benefits

This preprocessing reduces measurement noise across runs while preserving color representational geometry, thereby improving the quality of subsequent SRM fitting.

## S16. Comparison with Retinal-Family Distortion Models

Two retinal-family models were evaluated alongside the 2-component model by the same grid-search and permutation procedure (see Section ).

**Models.** The 1-DOF Machado model Machado et al., 2009 shifts the L/M cone spectral sensitivity by  $\Delta\lambda$  nm, yielding a hue-confusion pattern characteristic of the CVD subtype. The 2-DOF retinal+cortical R+C model Tregillus et al., 2021 augments



Machado with a cortical gain  $g$ :  $r'_g = r_{g,\text{base}} + (1 + g)(r_{g,\text{ret}} - r_{g,\text{base}})$ , where  $g = 0$  recovers the retinal-only model and  $g = -1$  is exact cortical compensation.

**Fits (Table 1).** Under the historical LOCO- $\rho$  argmax criterion (40,320 label permutations), reported here for reference only, retinal-family fits at hV4 yield Sub-08 R+C ( $\Delta\lambda = 2.0$  nm,  $g = +2.25$ ,  $\rho = 0.857$ ) and Sub-09 1-DOF Machado ( $\Delta\lambda = 13.5$  nm,  $\rho = 0.762$ ); the latter shift magnitude falls within the established range for moderate–severe protanomaly (Neitz & Neitz, 2011). The large positive cortical gain for Sub-08 ( $g = +2.25$ , indicating >100% overshoot beyond the exact Tregillus compensation level of  $g = -1$ ; Tregillus et al., 2021) is not physiologically interpretable as a gain parameter and is treated as an empirical characterization of residual hV4 structure. The cross-subject model-class divergence (R+C for Sub-08; 1-DOF Machado for Sub-09) prevents a common-axis comparison of the two cases’ distortion mechanisms.

**$\Delta$ RDM cross-model convergence.** Using R+C parameters for Sub-09 ( $\Delta\lambda = 19.5$  nm,  $g = -1.10$ ), an independent  $\Delta$ RDM test reaches  $p = 0.026$  (40,320 cosine-similarity permutations on crossnobis distances), providing cross-model corroboration for the Sub-09 distortion signal. Retinal-family optimization directly against  $\Delta$ RDM fails for both CVD subjects; LOCO is the more discriminating phenotype and  $\Delta$ RDM serves as a convergence cross-check only.

**Arc-compression and LOCO vulnerability.** At Sub-09’s severity, Machado arc-compression ( $\Delta\lambda = 13.5$  nm) collapses three hues (c4 green, c5 cyan, c6 blue) to a single perceived angle in opponent hue space. These three collapsed hues fall within Sub-09’s LOCO vulnerability cluster (c5/c6 cyan/blue), confirming that arc-compression eliminates correction capacity precisely where Sub-09 requires it most.

## S17. HC leave-one-out magnitude-anchor distribution

Because the production loss differs between participants (sub-08:  $\gamma_{\text{OY}} + L_{\text{RDM}}^{(V2)}$ ; sub-09:  $\gamma_{\text{all}} + L_{\text{RDM}}^{(V1)}$ ), the HC leave-one-out anchor was computed per participant under that participant’s loss. Each of the  $n = 7$  HC participants was refit as a pseudo-CVD case with the remaining HC pool as reference, yielding a distribution of fitted-parameter

magnitudes  $\|\hat{\beta}\|$ :  $30.5^\circ$  to  $58.1^\circ$  (mean  $49.1^\circ$ ) under the sub-08 loss, and  $23.4^\circ$  to  $55.5^\circ$  (mean  $35.7^\circ$ ) under the sub-09 loss. Both CVD estimates fall within their respective HC ranges (sub-08  $\|\hat{\beta}\| = 42.4^\circ$ ; sub-09  $\|\hat{\beta}\| = 24.1^\circ$ ), so parameter magnitude alone does not separate the CVD cases from HC; the distribution is reported as a descriptive magnitude anchor, not as a hypothesis test of CVD-versus-HC specificity.

### S18. Effect Sizes for Single-Case Comparisons

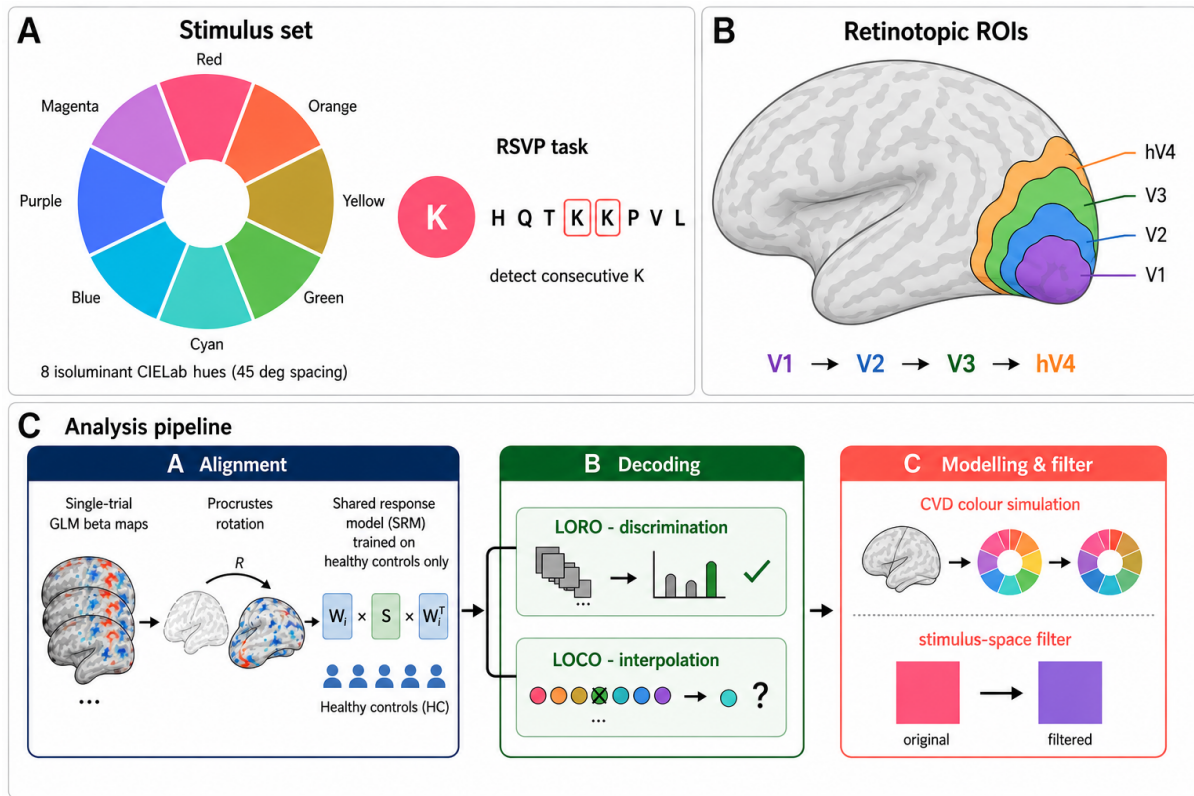
Crawford–Howell single-case comparisons used the modified  $t$  Crawford & Howell, 1998; the corresponding case-control effect size is  $d_{cc} = (x_{\text{case}} - \bar{x}_{\text{HC}})/s_{\text{HC}} = t \cdot \sqrt{(n+1)/n}$  where  $n$  is the HC sample size used in the test. Mann–Whitney  $U$  tests are reported with rank-biserial correlation  $r_{\text{rb}}$  as effect size. Full per-test statistics ( $t$ ,  $U$ ,  $p$ ,  $d_{cc}$ ,  $r_{\text{rb}}$ ) for every Crawford–Howell and Mann–Whitney comparison reported in the main text are tabulated in the online repository (`statistics/effect_sizes.csv`).

Table 1

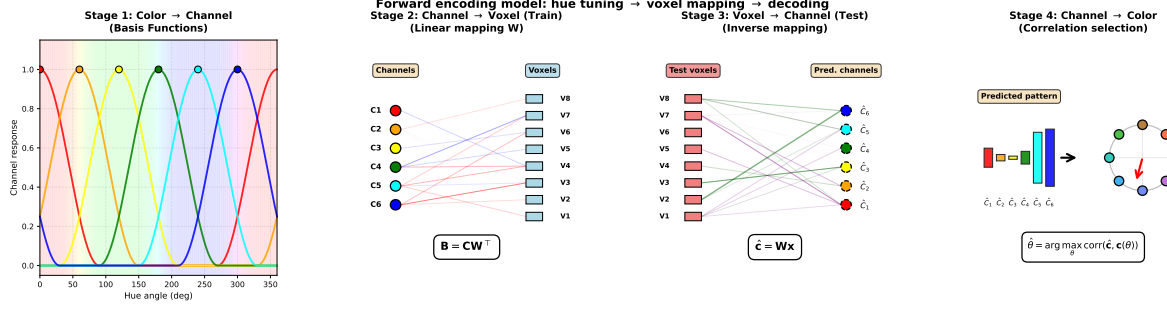
*Distortion-model fits at hV4 for both CVD participants. The 2-component model is listed at its adopted fit, selected by held-out composite test-loss ( $\bar{L}_{\text{test}}$ , lower is better; main text, Section ). The retinal-family baselines were fit only under the historical LOCO- $\rho$  argmax criterion (Section A.1 of Appendix A;  $\rho$  is Spearman correlation between simulated and observed per-color vulnerability), reported for reference only.*

| Subject         | Model                | DOF | Param. 1                          | Param. 2              | Fit                             |
|-----------------|----------------------|-----|-----------------------------------|-----------------------|---------------------------------|
| Sub-08 (deutan) | 2-component          | 2   | $\beta_s = 6^\circ$               | $\beta_c = -42^\circ$ | $\bar{L}_{\text{test}} = -2.36$ |
| Sub-09 (protan) | 2-component          | 2   | $\beta_s = 2^\circ$               | $\beta_c = +24^\circ$ | $\bar{L}_{\text{test}} = -1.54$ |
| Sub-08 (deutan) | R+C <sup>a</sup>     | 2   | $\Delta\lambda = 2.0 \text{ nm}$  | $g = +2.25$           | $\rho = 0.857$                  |
| Sub-09 (protan) | Machado <sup>a</sup> | 1   | $\Delta\lambda = 13.5 \text{ nm}$ | —                     | $\rho = 0.762$                  |

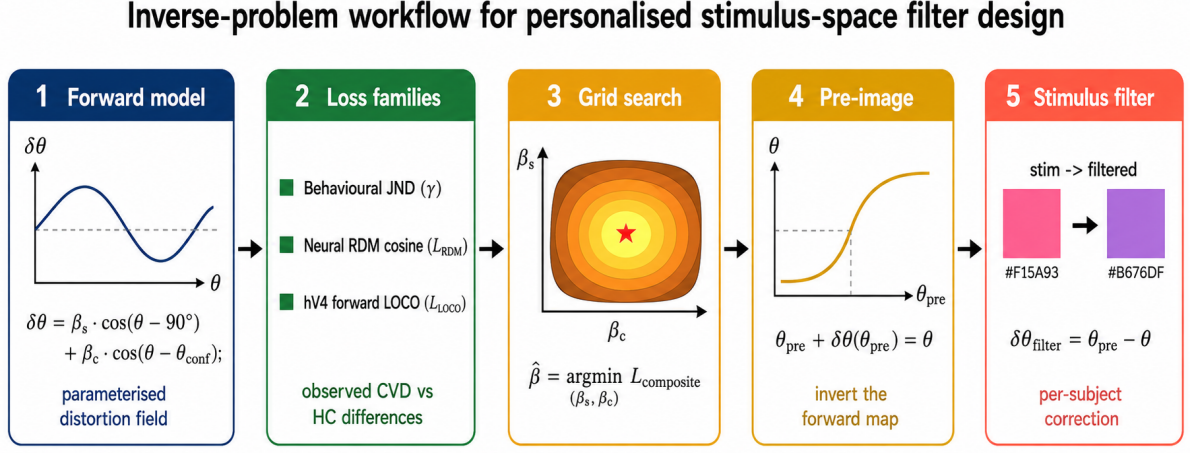
<sup>a</sup>Historical LOCO- $\rho$  argmax criterion (reference only); not the adopted filter fit.



**Figure 1. Experimental paradigm and analysis pipeline.** (A) Stimulus set: 8 isoluminant CIE  $L^*a^*b^*$  hues at  $45^\circ$  angular spacing ( $L^* = 75$ , chroma = 40; red–magenta). (B) Retinotopic ROIs: V1, V2, V3, hV4 (standard retinotopic mapping). (C) Analysis pipeline. Stage A: two-stage GLM single-trial  $\beta$  amplitudes, Procrustes-aligned across runs, projected into an SRM space trained on HC participants only. Decoding: LORO (discrimination), LOCO (interpolation). CVD characterization: 2-component cortical model with S-cone-axis ( $\beta_s$ ) and confusion-axis ( $\beta_c$ ) components. Pre-image yields per-subject stimulus-space filter. HC:  $n = 7$  (sub-01–07); deutan: sub-08; protan: sub-09.



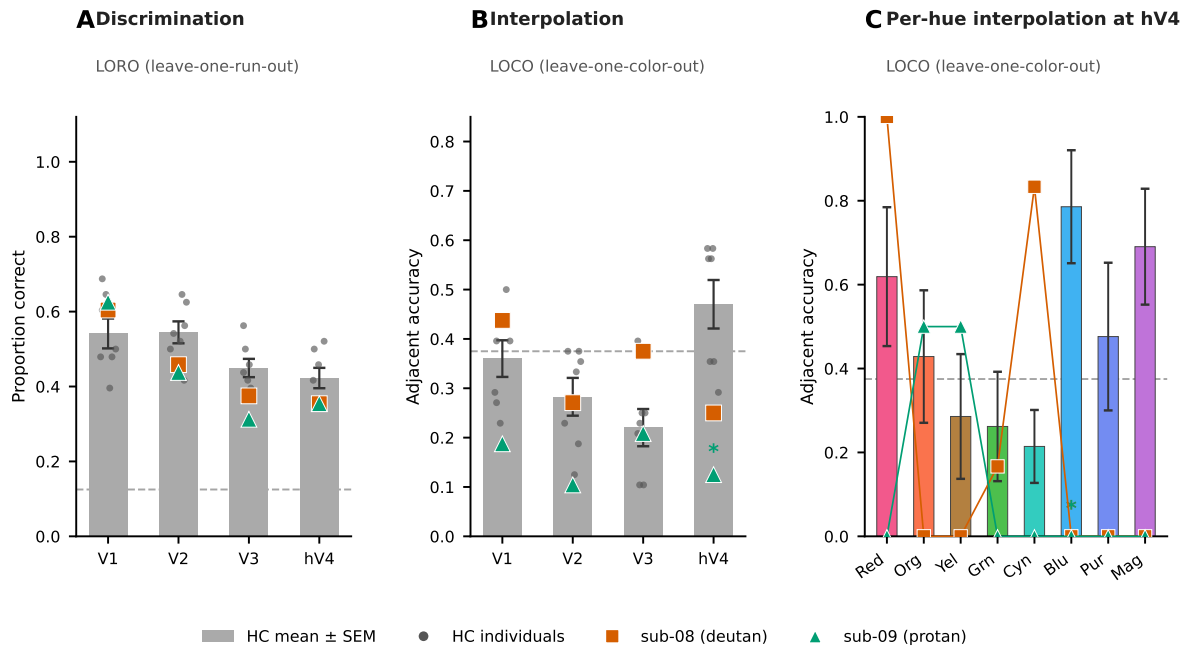
*Figure 2. Forward encoding model schematic.* Stage 1: each displayed hue is encoded as channel activations by six half-wave rectified squared-cosine basis functions, with preferred hues spaced at 60 degrees (Brouwer & Heeger 2009). Stage 2 (training): a per-voxel weight matrix  $W$  is fit by ridge regression with GCV-selected regularization, mapping channels to voxel responses. Stage 3 (testing): for a held-out voxel pattern, channel activations are reconstructed as  $\hat{\mathbf{c}} = W\mathbf{x}$ . Stage 4: the decoded hue is recovered as the angle whose basis vector has the highest Pearson correlation with the reconstructed channel pattern. The same encoder is used for cross-run color-discrimination decoding (Section ) and cross-color interpolation decoding (Section ); the test set differs but the basis, weights, and correlation-based selection are identical.



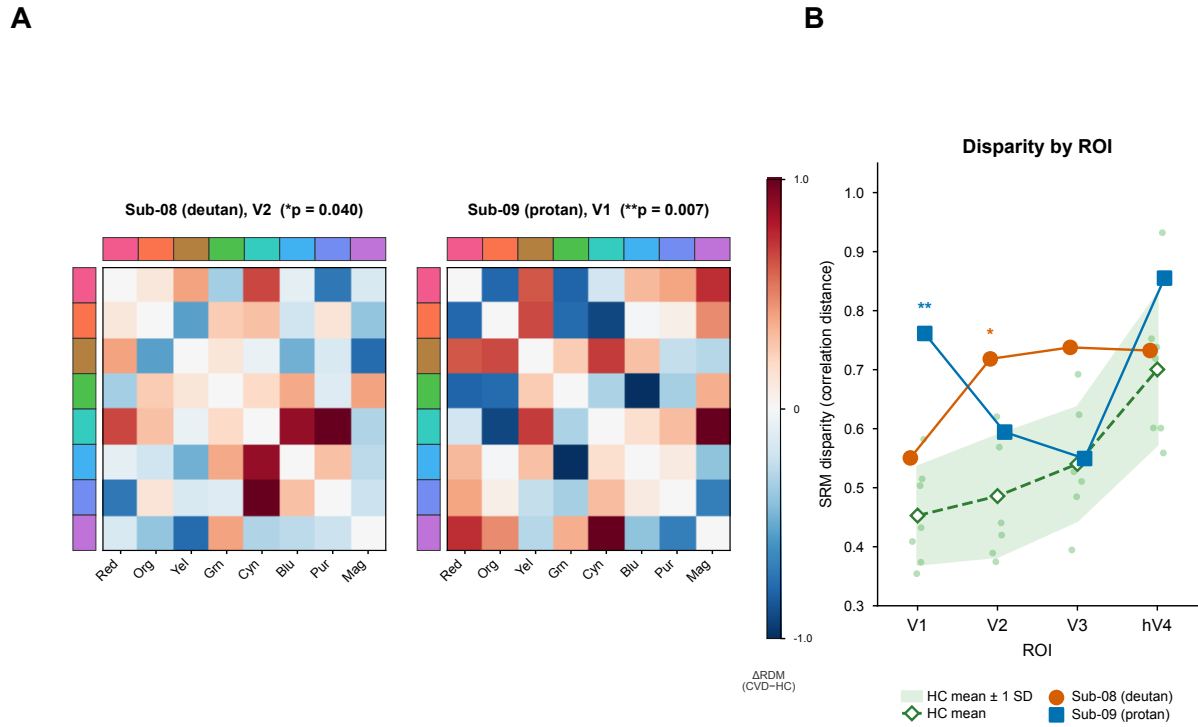
*Schematic; glyphs illustrative*

*Figure 3. Inverse-problem workflow for personalized stimulus-space filter*

**design.** (1) A two-parameter forward model expresses the per-hue distortion as  $\delta\theta(\theta) = \beta_s \cos(\theta - 90^\circ) + \beta_c \cos(\theta - \theta_{\text{conf}})$ : an S-cone-axis component ( $\beta_s$ ) plus a family-specific confusion-axis rotation ( $\beta_c$ ;  $\theta_{\text{conf}} = 16^\circ$  protan,  $150^\circ$  deutan). (2) Three loss families quantify CVD–HC differences — behavioural JND ratios ( $\gamma$ ), neural representational  $\Delta\text{RDM}$  cosine, and hV4 forward-encoding LOCO; only the LOCO loss couples to the forward encoder. (3) A grid search over  $(\beta_s, \beta_c)$  minimizes the  $z$ -scored composite to the selected fit  $\hat{\beta}$ . (4) The fitted forward map is analytically inverted: for each displayed hue  $\theta$ , the corrected stimulus  $\theta_{\text{pre}}$  solves  $\theta_{\text{pre}} + \delta\theta(\theta_{\text{pre}}) = \theta$ . (5) The per-subject filter is  $\delta\theta_{\text{filter}}(\theta) = \theta_{\text{pre}} - \theta$ . Glyphs are illustrative; the fitted loss landscape is shown in Figure 6.

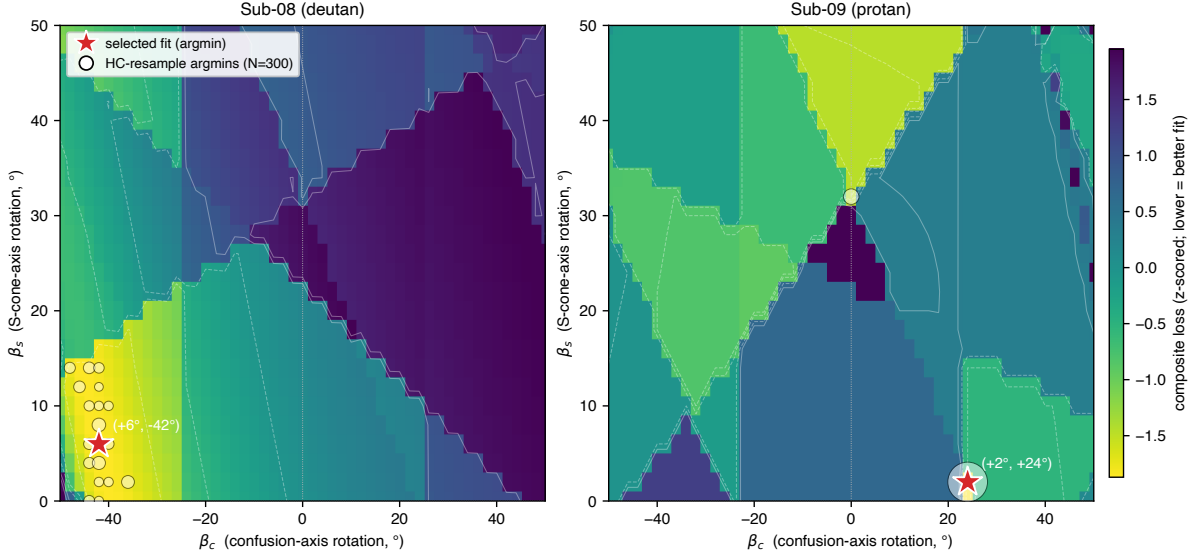


*Figure 4. Color discrimination is preserved but interpolation is selectively impaired at hV4 in CVD. (A)* LORO discrimination accuracy (forward encoding model, 8-class, SRM-aligned), V1–hV4. HC mean  $\pm$  SEM ( $n = 7$ ;  $n = 6$  at hV4, sub-07 excluded for low voxel count; gray); sub-08 (orange square), sub-09 (teal triangle). Chance =  $1/8$ . **(B)** LOCO adjacent accuracy ( $\pm 1$  hue step; chance =  $3/8$ ). \*: sub-09 below HC at hV4 (Crawford & Howell  $t = -2.48$ ,  $p = 0.024$ ; exploratory single-case finding); sub-08  $t = -1.58$ ,  $p = 0.082$ , n.s. **(C)** Per-hue adjacent accuracy at hV4, HC bars  $\pm$  SEM ( $n = 6$ ). Per-hue Crawford & Howell tests (one-tailed) were significant only for blue (both participants  $d = 2.20$ ,  $p = 0.042$ ); purple ( $d = 1.02$ ,  $p = 0.19$ ) and magenta ( $d = 1.89$ ,  $p = 0.064$ ) were n.s. Joint pattern in (A) and (B–C) defines the filter precondition and the functional vulnerability profile, respectively.

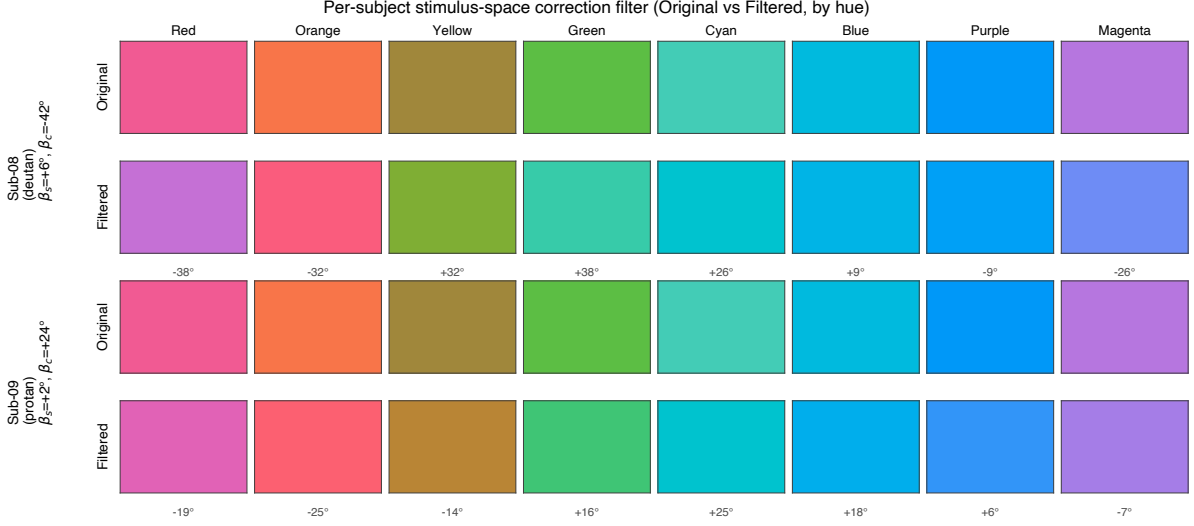


**Figure 5. Each CVD subject shows significantly elevated color representation disparity at a distinct ROI. (A)**  $\Delta RDM = RDM_{CVD} - \overline{RDM}_{HC}$  in SRM-aligned space at each subject's primary ROI (sub-08 V2, sub-09 V1). Warm: CVD distances larger than HC mean; cool: smaller. **(B)** Mean pairwise correlation distance (disparity) per subject per ROI. HC band: group mean  $\pm 1$  SD ( $n = 7$ ); individual HC LOO as dots. Crawford & Howell tests (sub-08 V2:  $p = 0.040^*$ ; sub-09 V1:  $p = 0.007^{**}$ , exploratory single-case). (A) shows the directional pattern of pairwise deviations. (B) tests the absolute amplitude of pairwise distances against the HC LOO distribution. Each subject showed elevated disparity at a single, distinct ROI (sub-08 V2; sub-09 V1).

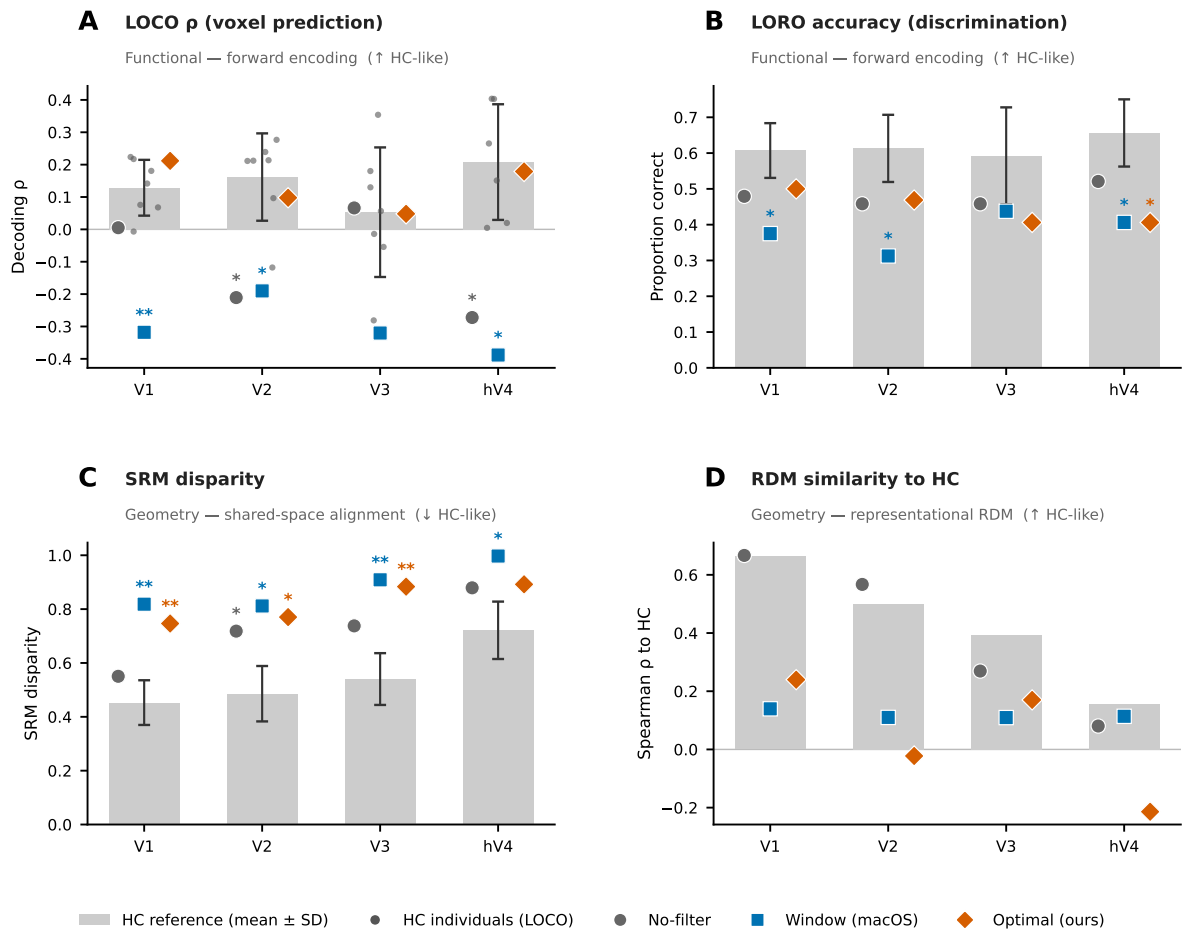




**Figure 6. Per-subject loss landscape and parameter uncertainty.**  $z$ -scored composite loss over the  $(\beta_s, \beta_c)$  grid for each CVD participant’s selected loss combination (sub-08:  $\gamma_{\text{OY}} + L_{\text{RDM}}^{(V2)}$ ; sub-09:  $\gamma_{\text{all}} + L_{\text{RDM}}^{(V1)}$ ), reconstructed on the full 7-HC pool. Lower (yellow) indicates better fit. Red star: selected fit (argmin) — sub-08  $(\hat{\beta}_s, \hat{\beta}_c) = (6^\circ, -42^\circ)$ ; sub-09  $(2^\circ, +24^\circ)$ . White points: per-resample argmins over  $N = 300$  HC 5-train/2-test resamples, showing parameter uncertainty. The selected fits are descriptive low-dimensional embeddings of each participant’s distortion, not physiological point estimates (this section).



**Figure 7. Per-subject stimulus-space correction filter.** For each CVD participant, the displayed stimulus (*Original*) and the corrected stimulus (*Filtered*) are shown for the eight scanner hues. Left block: sub-08 (deutan),  $(\hat{\beta}_s, \hat{\beta}_c) = (6^\circ, -42^\circ)$ . Right block: sub-09 (protan),  $(\hat{\beta}_s, \hat{\beta}_c) = (2^\circ, +24^\circ)$ . The filtered stimulus is the exact numerical pre-image of the fitted 2-component transform; the per-hue correction  $\delta\theta_{\text{filter}} = \theta_{\text{pre}} - \theta$  is annotated beneath each *Filtered* swatch (8/8 pre-images exact, residual  $< 0.01^\circ$  for both subjects). Rendering coordinate space: STIM\_LAB CIELab (project convention). Quantitative behavioural validation of filter efficacy is deferred to the Phase-3 2AFC arm and will be added once data are collected.



*Figure 8. The personalized filter restores HC-level forward-tuning but not an HC-like color geometry (sub-08, second session). Functional indices (A, B) and representational geometry (C, D) across V1–hV4. Gray bars, HC reference (mean  $\pm$  SD,  $n = 7$ ); dots in (A), individual HC values. Markers: no-filter (gray circle, Session-1 baseline), deployed macOS filter (blue square), personalized filter (orange diamond). (A) LOCO forward-tuning correlation to the canonical hue map ( $\uparrow$  HC-like). (B) LORO color-classification accuracy ( $\uparrow$  HC-like). (C) SRM disparity into the HC shared space ( $\downarrow$  HC-like). (D) Shared-space correlation-distance RDM similarity to HC ( $\uparrow$  HC-like). Single subject, descriptive. Stars, Crawford–Howell single-case deviation from the HC distribution ( $*p < .05$ ,  $**p < .01$ ,  $***p < .001$ ).*